Working Paper

Data Documentation

COVID-19 Fiscal Response Database: Codebook Version 1.7

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ABSTRACT

This codebook documents a comprehensive micro-level dataset of fiscal policy responses to the COVID-19 pandemic across all 38 OECD member states during 2020–2021. The dataset contains 2,301 fiscal measures including direct government interventions like subsidies, loans, guarantees or tax measures, with detailed information on timing, target groups, policy instruments, and fiscal commitments. Each measure is documented with primary source verification, enabling systematic cross-country comparison. The dataset distinguishes between announced and disbursed amounts. Each measure is further classified into one of three transmission channels—Demand Injection (DI), Capacity Preservation (CP), or Health (H)—based on a reproducible protocol operating at the `PolicyCode` level. Unlike existing databases that aggregate measures at higher levels, this micro-level structure enables novel research on policy heterogeneity, instrument choice, and compositional effects of fiscal packages between countries. The granular data allow researchers to analyze which specific policy instruments were deployed, when they were implemented, which groups were targeted, and how policy mixes evolved over time. This represents the first comprehensive micro-level fiscal policy cross-country database with complete source documentation and transmission channel classification for the pandemic period.

JEL Classification: E62, H12, H50, H60, I18

Keywords: COVID-19, fiscal policy, government spending, pandemic response, cross-country comparison, OECD

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1 Introduction

This codebook documents fiscal policy measures from a comprehensive dataset of government responses to the COVID-19 pandemic across all 38 OECD member states. The dataset provides detailed information on fiscal commitments through revenue and expenditure measures implemented during 2020–2021, offering a micro-level breakdown of all fiscal measures during this period. The dataset contains 4,683 total observations, of which 2,301 represent fiscal policy measures (`broad_fiscal` = 1, 2, or 3). The remaining 2,382 observations consist of monetary policy interventions documented separately. Each fiscal measure includes detailed descriptions, timing information, policy classification, target groups, and size of fiscal commitments in both absolute terms in local currency and as percentage of 2019 GDP.

1.1 Contribution and Scope

To the best of my knowledge, this represents the first micro-level dataset of fiscal responses during COVID-19 with comprehensive source documentation, enabling systematic cross-country comparison. Unlike existing datasets by the World Bank (Lacey et al., 2021) and IMF (Mishra et al., 2022), which aggregate measures at higher levels and cover only partial time periods, this database:

1. Disaggregates complex legislative packages into individual measures
2. Extends temporal coverage through 2021
3. Provides primary source documentation for each entry
4. Distinguishes between announced and effective/disbursed amounts
5. Includes detailed policy instrument classification

Remark: This codebook describes only fiscal measures. Monetary policy measures (`broad_fiscal=0`) are excluded from this documentation and will be addressed in a separate technical appendix.

2 Data Construction

2.1 Data Sources and Collection Process

The starting point for this database was the work by the World Bank (Lacey et al., 2021) and the IMF (Mishra et al., 2022) both of which attempted to create a database at this micro-level. Nevertheless, neither dataset included data beyond 2020 (World Bank only until September 2020 and IMF until December 2020). The measures were captured too broadly, as new laws consisting of numerous individual measures such as for example the CARES Act enacted on March 27, 2020 in the USA were recorded as single entries. This made it impossible to disaggregate measures by type and target group. Additionally, there were gaps in coverage, with some measures and entire countries missing.

To address these gaps and refinements, I used these two datasets as a starting point and collected measures country by country. To ensure comprehensive coverage, I began by consulting existing trackers: the IMF Policy Tracker (International Monetary Fund, 2020), the OECD COVID-19 Policy Tracker (Organisation for Economic Co-operation and Development , OECD), the Yale COVID-19 Financial Response Tracker (Yale Program on Financial Stability, 2020), and for European countries, the ESRB Policy Measures Tracker (European Systemic Risk Board, 2020) and the European Commission Coronavirus Response (European Commission, 2020). Subsequently, I visited individual government websites to verify and expand upon measures based on official communications and documents and added the primary source. In some cases, particularly for smaller countries, secondary sources such as newspaper articles or reports from auditors were used to verify measures when primary sources were unavailable or incomplete. Every measure is documented with at least one source, and multiple sources are provided when available. As this is a key distinguishing feature of the dataset, and recognizing that web links may become outdated over time, each measure description is as comprehensive as possible and supplemented with program names and relevant legislation. This documentation strategy enables future researchers to verify measures even when original URLs are no longer accessible.

2.2 Classification System

Measures are categorized using a modified classification system adapted from (Lacey et al., 2021). The system comprises 9 main categories with 49 sub-categories, allowing identification of: (1) measure type (revenue vs. expenditure), (2) IMF classification (above-the-line vs. below-the-line), (3) whether measures involve new money or acceleration/refund of existing resources, and (4) primary impact area (households, health system, broad economy, or

firm categories). The classification system exhibits strong reliability at the broad category level. Users should exercise caution with fine-grained sub-categories, as some measures span multiple classifications. For robust analysis, I recommend using broad categories. Complete classification details appear in Section 6.

In addition, each fiscal measure is assigned to one of three transmission channels—Demand Injection (DI), Capacity Preservation (CP), or Health (H)—based on a reproducible protocol operating at the `PolicyCode` level. This classification maps each of the 49 policy codes to the structurally distinct channel through which the measure affects the economy. The transmission channel classification is documented in Section 7.

2.3 Measurement Conventions

2.3.1 Disbursed vs. Announced Amounts

Where data are available, all values reflect actual amounts disbursed or utilized. The exception is guarantees and loan facilities, consistently reported at announced ceiling amounts. For these instruments, ceiling amounts were typically set significantly above expected utilization, as the announced ceiling represents the economically relevant metric. These amounts signal government commitment to economic actors, influencing market confidence and borrower behavior regardless of actual uptake. Moreover, actual utilization of guarantees proves difficult to measure accurately, making announced ceilings more reliable indicators of program scale.

2.3.2 Policy Extensions

Programs extended without changes in size or target groups are captured with `fiscal_size=0` and categorized as `broad_fiscal=2`. These extensions are included for completeness to document full policy response timelines.

2.3.3 Treatment of Supranational Measures

The dataset includes only national government policies, with two exceptions for EU member states:

EIB Pan-European Guarantee Fund: All 27 EU member states contributed to this €25 billion guarantee fund proportional to their EIB capital share. The fund enabled the EIB Group to mobilize up to €200 billion in additional financing, focusing on SMEs. Since contributions were proportional to economic size, these guarantees are

attributed to individual member states as national government guarantees in the same size.

NextGenerationEU/Recovery and Resilience Facility: This €723 billion recovery instrument provides grants and loans to EU member states for reforms and investments. The RRF operates on a performance basis: member states receive funding only upon achieving pre-agreed milestones. To avoid double-counting with national measures, since RRF funds reimburse member states for qualifying expenditures already undertaken, these amounts are recorded separately under `broad_fiscal=E`. This approach tracks EU-level fiscal support while distinguishing it from national measures.

2.3.4 Subnational Government Transfers

Transfers to local governments are, where possible, assigned to final purpose. This assignment proves straightforward in most cases, as local governments manage specific policy areas within their jurisdictions. Transfers specifically offsetting local government revenue losses are categorized accordingly (9 observations).

3 Data Quality and Validation

3.1 Validation Against IMF Policy Tracker

To validate dataset reliability, aggregate figures were calibrated against the official numbers from the IMF Policy Tracker (covering 2020 to October 2021) (International Monetary Fund, 2020), that is widely used as a reference in the literature. Some differences from IMF figures occur due to definitional approaches, exchange rate calculations, and data availability. Most importantly, the IMF tracks strictly nominal announced values across all categories, whereas this dataset distinguishes between announced values (for guarantees and loans) and effective or disbursed values (for other fiscal measures where such data are available). Since the IMF does not provide detailed source documentation for individual measures, exact replication of their figures is neither feasible nor necessary given the methodological refinements applied in this dataset.

Deviations are measured in percentage points (pp) rather than percentage differences, as this provides consistent economic interpretation across countries. I categorize deviations as: excellent agreement (< 1 pp), good agreement (1–2 pp), acceptable deviation (2–5 pp), and significant deviation (> 5 pp).

3.2 Validation Results

Table 5 summarizes validation results across all 38 countries. Key findings:

- 22 countries (57.9%) show excellent agreement (< 1 pp)
- 32 countries (84.2%) show excellent/good agreement (≤ 2 pp)
- 36 countries (94.7%) fall within acceptable deviation threshold (≤ 5 pp)
- Mean absolute deviation: 1.4 percentage points
- Median absolute deviation: 0.7 percentage points

Where deviations exceed 2 pp, detailed justifications based on primary sources are documented in Appendix B. The two countries exceeding 5 pp (Germany, -8.3 pp; Chile, +9.2 pp) reflect more comprehensive coverage in this dataset, as IMF figures appear to undercount major programs documented in official government reports.

3.3 LLM-Assisted Quality Checks

To further strengthen the reliability, all entries underwent afterwards systematic quality checks supported by automated procedures. A large language model (LLM) with web browsing capabilities served as a final check assistive tool to flag potential inconsistencies between recorded measures and publicly available sources, as well as typographical errors, currency misspecifications, and data entry errors. The LLM systematically cross-referenced each entry's description, source, and reported amount against accessible documentation.
Flagged entries were manually reviewed and corrected where necessary, with all final coding decisions made by the author. This semi-automated approach enabled efficient verification of all entries while maintaining human oversight of data quality. LLM verification is limited to publicly accessible sources and cannot detect errors originating in original government documentation itself.

4 Dataset Description

4.1 Overview

The dataset `fiscal_classified_v1_7.xlsx` contains 4,683 observations across 38 OECD member states during 2020–2021. The complete dataset includes:

- 1,829 fiscal policy measures (`broad_fiscal=1`): Direct government spending, subsidies, grants, tax measures
- 443 policy extensions (`broad_fiscal=2`): Extensions without changes in size or target groups
- 29 EU-level measures (`broad_fiscal=3`): NextGenerationEU/RRF funding
- 2,382 monetary policy measures (`broad_fiscal=0`): Central bank interventions (documented separately)

This codebook documents fiscal-related entries only (`broad_fiscal=1, 2, 3`), totaling 2,301 observations.

4.2 Temporal and Geographic Coverage

The fiscal measures primarily cover 2020 (1,766 observations) and 2021 (535 observations), with highest concentration in Q1 (525 obs) and Q2 (723 obs) of 2020, reflecting initial government responses to the pandemic. Geographic coverage includes all 38 OECD member states. Table 4 presents the distribution of observations by country. European countries account for 1,624 observations, with substantial representation from Latin America & Caribbean (267 obs), East Asia & Pacific (215 obs), North America (120 obs), and Middle East & North Africa (75 obs). Advanced economies represent 1,884 observations (81.9%), while emerging markets account for 417 observations (18.1%).

4.3 Data Quality Considerations

Missing Values: The main part of the dataset (`broad_fiscal=1`) contains 5 missing values in fiscal size measurements (`broad_fiscal_size`). This reflects situations where fiscal commitment amounts were not publicly disclosed or quantified.
Cross-Country Comparability: The `broad_fiscal_gdp` variable enables cross-country comparisons by normalizing fiscal measures to GDP.
Multiple Recordings: Some policy measures may be recorded multiple times if announced in phases or modified over time. Users should consider this when trying to count the number of measures.

5 Variable Definitions

Table 1 provides complete variable definitions and descriptions for all variables in the dataset.

Table 1: Variable Definitions and Descriptions

Variable	Type	Description
ID	String	Unique identifier for each policy measure (format: CovidID-XXXX)
Country	String	Three-letter ISO country code (e.g., AUS, DEU, USA)
ifs	Numeric	International Financial Statistics country code
income_group	String	Economic classification: AE (Advanced Economies) or EM (Emerging Markets)
region	String	Geographic region (e.g., "Europe & Central Asia", "North America")
NGDP_2019	Numeric	Nominal GDP in 2019 (national currency, millions), used for normalization
Year	Numeric	Year when the measure was announced or implemented (2020 or 2021)
Quarter	Numeric	Quarter of announcement (1–4)
Month	Numeric	Month of announcement (1–12)
date	String	Exact date of announcement (DD-MM-YYYY)
description	Text	Detailed description of the policy measure, including program names, target groups, and implementation details
link	Text	Primary source URL(s) documenting the measure; multiple sources separated by semicolons
PolicyCode	Mixed	Specific policy instrument code (49 codes + general); see Table 2
PolicyCategory	Numeric	Main policy category (1–9); see Table 2
broad_fiscal	Mixed	Fiscal measure classification: 1 = Fiscal policy measure; 2 = Policy extension; 3 = EU-level measure; 0 = Monetary policy (excluded)
broad_fiscal_size	Numeric	Fiscal commitment amount in national currency (millions). For guarantees/loans: announced ceiling; for other measures: disbursed/utilized amount where available
broad_fiscal_gdp	Numeric	Fiscal commitment as percentage of 2019 GDP (broad_fiscal_size / NGDP_2019 × 100)
category	Numeric	IMF fiscal accounting: 1 = above-the-line; 2 = below-the-line; 3 = tax deferrals
transmission_channel	String	Transmission channel classification: DI (Demand Injection), CP (Capacity Preservation), or H (Health). Assigned at the `PolicyCode` level per the protocol in Section 7
is_extension	Binary	1 if `broad_fiscal` = 2 (policy extension with zero fiscal weight)
is_eu	Binary	1 if `broad_fiscal` = 3 (supranational EU-level measure)
is_deferral	Binary	1 if IMF `category` = 3 (tax deferral; classified as CP but structurally distinct)
is_blw	Binary	1 if IMF `category` = 2 (below-the-line: loans and guarantees)

6 Policy Classification System

6.1 Policy Category Classification

Table 2 lists all 49 policy codes, based on a modified version of (Lacey et al., 2021). This classification system disaggregates complex legislative packages into individual policy instruments. The 49-code taxonomy distinguishes between revenue and expenditure measures, identifies target groups, and captures instrument types.

Table 2: Policy Measure Codes

1 Revenue Measures to protect businesses	
1	Accelerated asset depreciation (CIT)
2	Extend loss carry-forward (CIT)
3	Broaden tax deductibility
4	Tax credits
5	Deferral of tax filing
6	Deferral of tax payments
7	Tax rate reduction
8	Tax amnesty / incentives
9	Accelerating refunds
10	Lower advance payment
11	Suspend debt collection
12	Suspend audit activities
13	Tax exemption
14	Tax waiver / suspension
2 Revenue Measures to protect individuals	
15	Deferral of tax filing
16	Deferral of tax payments
17	Tax rate reduction
18	Tax amnesty / incentives
19	Broaden tax deductibility
20	Tax credits
21	Tax exemption
22	Tax waiver / suspension
23	Tax-free provision for medical supplies
24	Customs duty relief on medical goods
3 Revenue Measures to boost consumption / demand	
25	Exemption of tax rates on consumption VIT
26	Lower tax rates on consumption VAT
4 Expenditure Measures to boost consumption / demand	
27	Investment into infrastructure
28	Green investments
29	Reactivate local tourism
5 Health expenditure measures	
31	Supply of low-cost medical items
32	Supply of high-cost medical items
33	Targeted infrastructure investments
34	Expansion and support of human resources
general	General health related spending
5.5 Revenue Measures to promote availability of medical items	
30	Lower tax rates for medical items
6 Expenditure Measures to individuals	
35	Direct cash transfers for individuals
36	Expansion of unemployment benefits
37	Temporary expansion of existing benefits
38	Supplementary ad hoc programs
39	Wage compensation / enhanced paid leave
7 Credit and equity measures	
40	Preferential loans to firms
41	Equity or quasi-equity investments
42	Preferential loans to households
43	Guarantees for loans for business
8 Expenditure Measures for businesses	
45	Income support general
46	Income support non-profit
47	Income support specific
48	Income support education sector
9 Transfers to local governments for losses	
49	Income support for local governments

6.2 IMF Fiscal Accounting Classification

While the Policy Category classification identifies what instruments were deployed and who was targeted, the IMF fiscal accounting framework captures how these measures affect government finances and balance sheets.

- **Category 1 – Above-the-Line:** Expenditure measures directly impacting government fiscal balance through revenue changes or expenditures. Examples include direct spending programs, wage subsidies, and tax reductions. These measures affect the government's net lending/borrowing position.
- **Category 2 – Below-the-Line:** Expenditure loans and guarantees that primarily affect liquidity and timing without immediate accrual impact on fiscal balance. While these create contingent liabilities, they do not directly affect the deficit at announcement. Examples include loan programs and credit guarantees.
- **Category 3 – Tax Deferrals:** Revenue timing measures allowing businesses or households to defer tax payments. These affect cash flow and liquidity but do not necessarily imply permanent revenue loss, as deferred amounts are typically paid at a later date.

7 Transmission Channel Classification

This section documents the classification of each fiscal policy measure into one of three transmission channels based on the theoretical framework developed in the accompanying paper. The classification operates at the `PolicyCode` level, ensuring reproducibility and eliminating subjective judgment at the individual measure level.

7.1 Classification Principle

The three transmission channels correspond to structurally distinct mechanisms through which fiscal interventions affect the economy:

- **Demand Injection (DI):** Measures that directly increase the flow of disposable income to households or aggregate expenditure in the economy. These transmit through the multiplier channel, shifting the output level by a constant amount per unit deployed.
- **Capacity Preservation (CP):** Measures that preserve productive structures—firm–worker matches, firm solvency, supply chain integrity—thereby reducing the persistence of the output gap. These counteract lockdown-induced hysteresis by maintaining the economy's productive capacity during the crisis.
- **Health (H):** Pandemic-specific health expenditures that are not a policy instrument in the fiscal composition problem but an endogenous cost driven by the epidemiological state. These provide the empirical basis for calibrating health-related cost parameters.

7.2 Decision Rules

The following hierarchy governs classification decisions when a measure's transmission channel is not immediately apparent from its `PolicyCode` definition:

1. **Primary criterion: transmission mechanism.** The decisive question is *how* the measure affects the economy, not *who* receives the funds. A wage subsidy paid to firms that maintains employment relationships is CP even though it ultimately supports worker income. A direct cash transfer to households is DI even if the household uses it to pay rent to a business.
2. **Recipient as tiebreaker.** When the transmission mechanism is ambiguous, the recipient disambiguates: measures targeting firms, employers, or productive structures default to CP; measures targeting households, individuals, or consumers default to DI.

3. **Tax deferrals.** All tax deferrals (IMF Category 3) are classified as CP regardless of whether the recipient is a business or individual. Deferrals operate as zero-interest short-term government loans that preserve liquidity and prevent insolvency cascades. They generate no permanent change in disposable income (the deferred amount is repaid) and thus have no sustained multiplier effect. Their primary transmission is capacity preservation—preventing liquidity-constrained agents from defaulting on obligations during the acute crisis phase.

4. **Health expenditures.** All measures classified under PolicyCategory 5 (Health expenditure measures) and 5.5 (Revenue measures to promote availability of medical items), as well as Codes 23 and 24, are assigned to H regardless of whether they involve expenditure or revenue instruments.

5. **Code-level homogeneity.** Classification is applied uniformly within each `PolicyCode`. For codes with heterogeneous content (notably Code 39), the dominant transmission mechanism determines the classification.

7.3 Complete Classification Mapping

Table 3 provides the exhaustive mapping from `PolicyCode` to transmission channel, organized by `PolicyCategory`.

Table 3: Classification of PolicyCodes to Transmission Channels

Code	Description	Channel	Rationale
Category 1: Revenue Measures to Protect Businesses			
1	Accelerated asset depreciation (CIT)	CP	Reduces firm tax burden; preserves cash flow for operations
2	Extend loss carry-forward (CIT)	CP	Allows firms to offset losses against prior profits; preserves solvency
3	Broaden tax deductibility	CP	Reduces effective tax rate on firms; preserves operating margin
4	Tax credits (business)	CP	Reduces firm tax liability; preserves cash flow

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Code	Description	Channel	Rationale
5	Deferral of tax filing (business)	CP	Liquidity preservation (Decision Rule 3)
6	Deferral of tax payments (business)	CP	Liquidity preservation (Decision Rule 3)
7	Tax rate reduction (business)	CP	Reduces structural cost burden on firms; preserves firm viability
8	Tax amnesty / incentives (business)	CP	Removes penalty-driven insolvency risk
9	Accelerating refunds (business)	CP	Injects liquidity to firms from already-owed government obligations
10	Lower advance payment	CP	Reduces prepayment burden; liquidity preservation
11	Suspend debt collection	CP	Prevents enforcement-driven firm failure
12	Suspend audit activities	CP	Administrative relief; reduces compliance burden during crisis
13	Tax exemption (business)	CP	Eliminates specific tax obligations; preserves firm cash flow
14	Tax waiver / suspension (business)	CP	Includes bankruptcy moratoriums; prevents firm exit
Category 2: Revenue Measures to Protect Individuals			
15	Deferral of tax filing (individual)	CP	Liquidity preservation (Decision Rule 3)
16	Deferral of tax payments (individual)	CP	Liquidity preservation (Decision Rule 3)
17	Tax rate reduction (individual)	DI	Permanent increase in disposable income; standard multiplier channel
18	Tax amnesty / incentives (individual)	DI	Increases disposable income of households
19	Broaden tax deductibility (individual)	DI	Increases disposable income via lower effective tax rate
20	Tax credits (individual)	DI	Direct increase in household disposable income

Continued on next page

Code	Description	Channel	Rationale
21	Tax exemption (individual)	DI	Eliminates tax obligation; raises disposable income
22	Tax waiver / suspension (individual)	DI	Reduces household tax burden; increases disposable income
23	Tax-free provision for medical supplies	H	Health-adjacent; facilitates medical supply availability
24	Customs duty relief on medical goods	H	Health-adjacent; facilitates medical supply availability
Category 3: Revenue Measures to Boost Consumption/Demand			
25	Exemption of consumption tax rates	DI	Reduces consumer prices; stimulates aggregate expenditure
26	Lower consumption tax rates (VAT)	DI	Reduces consumer prices; transmits through spending channel
Category 4: Expenditure Measures to Boost Consumption/Demand			
27	Investment into infrastructure	DI	Government spending multiplier; direct demand injection
28	Green investments	DI	Government spending multiplier; direct demand injection
29	Reactivate local tourism	DI	Consumption stimulus targeting specific sectors
Category 5: Health Expenditure Measures			
30	Lower tax rates for medical items	H	Health system cost
31	Supply of low-cost medical items	H	Health system cost
32	Supply of high-cost medical items	H	Health system cost
33	Targeted health infrastructure	H	Health system cost
34	Expansion of human resources (health)	H	Health system cost
gen.	General health related spending	H	Health system cost
Category 6: Expenditure Measures to Individuals			
35	Direct cash transfers for individuals	DI	Canonical demand injection; direct income flow to households

Continued on next page

Code	Description	Channel	Rationale
36	Expansion of unemployment benefits	DI	Increases disposable income of unemployed; multiplier channel
37	Temporary expansion of existing benefits	DI	Augments existing transfer programs; income flow to households
38	Supplementary ad hoc programs	DI	Pandemic-specific income support to individuals
39	Wage compensation / enhanced paid leave	CP	Dominant mechanism: preserves employer–employee matches (Kurzarbeit, CJRS, JobKeeper, Activité Partielle, NOW). See Section 7.4.1.
Category 7: Credit and Equity Measures			
40	Preferential loans to firms	CP	Preserves firm solvency through liquidity provision
41	Equity or quasi-equity investments	CP	Preserves firm capital structure; prevents insolvency
42	Preferential loans to households	DI	Provides liquidity to households; transmits through spending
43	Guarantees for loans for business	CP	Preserves firm access to credit; prevents insolvency cascade
Category 8: Expenditure Measures for Businesses			
45	Income support general (business)	CP	Preserves firm operations; prevents exit of viable businesses
46	Income support non-profit	CP	Preserves organizational capacity of non-profit sector
47	Income support specific (sectoral)	CP	Preserves sector-specific productive capacity
48	Income support education sector	CP	Preserves institutional capacity in education
Category 9: Transfers to Local Governments			
49	Income support for local governments	CP	Preserves public service delivery capacity; prevents layoffs

7.4 Treatment of Special Cases

7.4.1 Code 39: Wage Compensation / Enhanced Paid Leave

Code 39 is classified as CP at the code level. Among the 159 measures coded as 39 across 37 countries, the overwhelming majority (>90% by fiscal volume) are short-time work schemes (Kurzarbeit, CJRS, Activité Partielle, NOW, JobKeeper) and apprenticeship wage subsidies—instruments whose defining feature is that the payment flows *through* the employer to maintain the employment relationship. A small number of entries (e.g., parental income compensation for school closures, pandemic leave payments) are structurally closer to DI, as they compensate individual income loss without preserving a productive firm–worker match.

Code-level classification is chosen over case-by-case assignment for three reasons. First, it ensures reproducibility: any researcher applying the protocol to the same dataset will obtain identical results. Second, the misclassification bias is bounded: the DI-like entries within Code 39 represent less than 10% of the code's fiscal volume. Third, the sensitivity of results to this decision can be tested by reclassifying the flagged entries.

7.4.2 Code 45: Income Support General (Business)

Code 45 (326 observations) contains predominantly firm-level grants and income support: German Soforthilfe, French Fonds de Solidarité, and similar programs across OECD economies. A small number of entries involve intergovernmental transfers with mixed purpose. These are classified as CP at the code level, consistent with the code's definition under Category 8 (Expenditure Measures for Businesses).

7.4.3 Tax Deferrals (IMF Category 3)

All tax deferral measures are classified as CP regardless of the recipient (business or individual). This reflects the economic nature of deferrals as temporary liquidity injections: the government provides an implicit zero-interest loan that must be repaid. The transmission mechanism is capacity preservation (preventing liquidity-driven default), not demand injection (no permanent increase in disposable income). Note that this classification implies that the effective fiscal cost for the deferral component of CP is substantially below the weighted average, since the realized fiscal cost of deferrals approaches zero for solvent taxpayers. Sensitivity analysis should examine results under exclusion of deferrals from CP.

7.4.4 Health Expenditures

Health measures (Codes 23, 24, 30–34, and `general`; approximately 0.43% of GDP) are excluded from both DI and CP. In the theoretical model, these expenditures are not a choice variable in the planner's fiscal composition problem but an endogenous cost driven by the epidemiological state.

7.4.5 Policy Extensions (`broad_fiscal` = 2)

Policy extensions carry `broad_fiscal_size` = 0 and document the temporal extension of existing programs without additional fiscal commitment. Each extension inherits the transmission channel of its base `PolicyCode`. Extensions are flagged (`is_extension` = 1) so they can be excluded from volume calculations while preserving timeline information.

7.4.6 EU-Level Measures (`broad_fiscal` = 3)

The 29 NextGenerationEU/RRF entries are classified based on their `PolicyCode` but carry a separate flag (`is_eu` = 1). These supranational measures differ from national fiscal responses in timing (disbursement lagged), financing (EU-level debt), and conditionality (milestone-based). They should be analyzed separately or excluded from the baseline specification.

7.4.7 Code `other`

The residual category `other` (41 observations, all policy extensions with zero fiscal weight) is classified case-by-case based on the measure description. Since all entries have `broad_fiscal_gdp` = 0, misclassification has no quantitative effect on the empirical analysis.

7.5 Robustness Flags

The classification adds the following binary flags for sensitivity analysis:

- `is_extension`: 1 if `broad_fiscal` = 2 (zero fiscal weight).
- `is_eu`: 1 if `broad_fiscal` = 3 (supranational measure).
- `is_deferral`: 1 if IMF `category` = 3 (tax deferral).
- `is_blw`: 1 if IMF `category` = 2 (below-the-line: loans and guarantees).

These flags enable the following robustness checks:

1. **Exclude deferrals from CP**: Reclassify `is_deferral` = 1 measures as a separate category or exclude entirely, testing whether the CP effectiveness parameter is sensitive to the inclusion of pure timing measures.

2. **Exclude extensions**: Restrict analysis to `broad_fiscal` = 1 (core fiscal measures with positive fiscal weight).

3. **Exclude EU measures**: Restrict to national fiscal responses.

4. **Separate above- and below-the-line CP**: Partition CP into direct measures (grants, subsidies; `is_blw` = 0) and contingent measures (loans, guarantees; `is_blw` = 1), testing whether effectiveness differs across sub-categories.

7.6 Validation

The classification can be cross-validated against two external benchmarks:

1. **IMF fiscal accounting**: Above-the-line measures (`category` = 1) should appear predominantly in both DI and CP; below-the-line measures (`category` = 2) should be exclusively CP (loans and guarantees preserve firm capacity). Tax deferrals (`category` = 3) should be exclusively CP per the protocol.

2. **Aggregate consistency**: The country-level DI and CP aggregates (as % of GDP) should be broadly consistent with the IMF's decomposition of above-the-line versus below-the-line spending, adjusted for the transmission-based reclassification of certain above-the-line measures.

8 Data Availability and Citation

8.1 Access

8.2 Recommended Citation

8.3 Version History

- **Version 1.7** (19. March 2026): Integrated transmission channel classification (DI/CP/H); added `transmission_channel` and robustness flag variables; added Codes 23/24; fixed PolicyCode misclassifications via `fix_classifications_v2.R`;
  updated to `fiscal_classified_v1_7.xlsx`
- **Version 1.3** (20. November 2025): Added IMF deviations appendix
- **Version 1.2** (19. November 2025): Updated codebook with enhanced documentation
- **Version 1.0** (13. November 2025): Initial version

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A Country Coverage and Observation Distribution

Table 4 presents the complete distribution of fiscal policy observations by country.

Table 4: Distribution of Fiscal Observations by Country

Country	Observations	Income Group	Region
Colombia	109	EM	Latin America & Caribbean
Germany	99	AE	Europe & Central Asia
Greece	98	AE	Europe & Central Asia
Ireland	93	AE	Europe & Central Asia
Spain	90	AE	Europe & Central Asia
United Kingdom	82	AE	Europe & Central Asia
Chile	78	EM	Latin America & Caribbean
France	78	AE	Europe & Central Asia
Israel	75	AE	Middle East & North Africa
Belgium	73	AE	Europe & Central Asia
Portugal	73	AE	Europe & Central Asia
Finland	73	AE	Europe & Central Asia
Italy	70	AE	Europe & Central Asia
USA	65	AE	North America
Hungary	65	AE	Europe & Central Asia
Australia	63	AE	East Asia & Pacific
Japan	57	AE	East Asia & Pacific
Austria	55	AE	Europe & Central Asia
Canada	55	AE	North America
Denmark	55	AE	Europe & Central Asia
Slovenia	53	AE	Europe & Central Asia
Norway	52	AE	Europe & Central Asia
South Korea	52	AE	East Asia & Pacific
Czech Republic	51	AE	Europe & Central Asia
Lithuania	50	AE	Europe & Central Asia
Netherlands	50	AE	Europe & Central Asia
Sweden	49	AE	Europe & Central Asia
Estonia	48	AE	Europe & Central Asia
Iceland	48	AE	Europe & Central Asia
Turkey	44	EM	Middle East & North Africa
New Zealand	43	AE	East Asia & Pacific
Poland	41	AE	Europe & Central Asia
Costa Rica	40	EM	Latin America & Caribbean
Mexico	40	EM	Latin America & Caribbean
Luxembourg	36	AE	Europe & Central Asia
Switzerland	34	AE	Europe & Central Asia
Latvia	33	AE	Europe & Central Asia
Slovakia	31	AE	Europe & Central Asia
Total	**2,301**		

Notes: Observation counts include all fiscal measures (broad_fiscal = 1, 2, 3). Countries ordered by number of observations.

B IMF Policy Tracker Comparison: Detailed Country Results

This appendix presents detailed country-by-country comparisons between this dataset and the IMF Policy Tracker. Deviations are measured in percentage points (pp) of GDP.

B.1 Methodology

Deviations are categorized as:

- **Excellent agreement** (< 1 pp): Minor rounding and exchange rate differences
- **Good agreement** (1–2 pp): Acceptable methodological differences
- **Acceptable deviation** (2–5 pp): Requires justification based on primary sources
- **Significant deviation** (> 5 pp): Requires detailed analysis and documentation

B.2 Summary of Validation Results

Table 5: IMF Validation Results by Agreement Level

Agreement Level	Countries	Percentage	Avg. Deviation
Excellent (< 1 pp)	22	57.9%	0.4 pp
Good (1–2 pp)	10	26.3%	1.3 pp
Acceptable (2–5 pp)	4	10.5%	3.2 pp
Significant (> 5 pp)	2	5.3%	9.1 pp
Total	**38**	**100.0%**	**1.4 pp**

Notes: Mean absolute deviation across all countries: 1.4 pp. Median absolute deviation: 0.7 pp. 84.2% of countries show deviations ≤2 pp; 94.7% show deviations ≤5 pp.

B.3 Significant Deviations (> 5 pp)

Germany (IMF: 44.1% — Database: 35.8% — Deviation: -8.3 pp) The deviation reflects the IMF's overstatement of guarantee programs at €826 billion versus the actual €400 billion documented in official federal government documentation.

Chile (IMF: 16.9% — Database: 26.1% — Deviation: +9.2 pp) The large discrepancy is driven by broader coverage in this dataset. In particular, the IMF figures largely exclude the guarantees that were announced. Official documents clearly indicate that guarantee schemes amounted to around 10% of GDP. If this guaranteed component is subtracted from my totals, the resulting figures align closely with the IMF numbers official budget documentation.

B.4 Acceptable Deviations (2 – 5 pp)

Estonia (IMF: 10.9% — Database: 15.8% — Deviation: +4.9 pp) The IMF documentation completely omits tax deferral measures totaling approximately 5 pp of GDP. These measures are clearly documented in official government sources, including parliamentary budget documents. When excluding tax deferrals, both figures align closely at approximately 11% of GDP.

Portugal (IMF: 12.1% — Database: 14.9% — Deviation: +2.8 pp) Differences arise from incorrect guarantee valuations in the IMF documentation. When adjusted to reflect actual guarantee ceiling amounts as documented in European Commission state aid approvals, my figures align at approximately 12.5% of GDP.

Sweden (IMF: 16.0% — Database: 13.5% — Deviation: -2.5 pp) The IMF substantially overestimates tax deferrals by approximately SEK 200 billion while underestimating guarantee envelopes by about SEK 150 billion. The Swedish National Audit Office confirms actual tax deferral utilization was significantly lower than announced ceilings.

Turkey (IMF: 16.7% — Database: 14.2% — Deviation: -2.5 pp) This dataset excludes quasi-fiscal operations (noncommercial activities of public corporations on behalf of government) as these do not constitute direct government spending nor carry explicit government guarantees.

B.5 Good Agreement (1–2 pp)

Eleven countries show deviations between 1–2 pp: Ireland (-1.8 pp), New Zealand (+1.7 pp), Australia (-1.5 pp), Japan (-1.3 pp), Austria (-1.2 pp), Germany (+1.1 pp), France (+1.1 pp), United Kingdom (+1.1 pp), Canada (-1.1 pp), Colombia (-1.1 pp) and Switzerland (-1.1 pp). These deviations primarily reflect exchange rate variations, minor temporal boundary differences, and the distinction between announced versus disbursed amounts.

B.6 Excellent Agreement (< 1 pp)

Twenty-two countries show deviations below 1 pp, reflecting excellent concordance with IMF figures: Hungary, Spain, Finland, United States, Greece, Italy, Iceland, Korea, Luxembourg, Belgium, Lithuania, Denmark, Norway, Slovenia, Slovak Republic, Costa Rica, Poland, Netherlands, Czech Republic, Israel, Latvia, and Mexico. These minor differences are attributable to standard exchange rate variations and rounding differences.

B.7 Interpretation

The high degree of concordance (84% within 2 pp; 97% within 5 pp) validates the comprehensiveness and accuracy of this dataset. Where significant deviations exist, they are systematically documented with primary source verification. The mean deviation of 1.2 pp and median of 0.7 pp indicate that typical differences are well within acceptable bounds for cross-country fiscal policy comparisons.